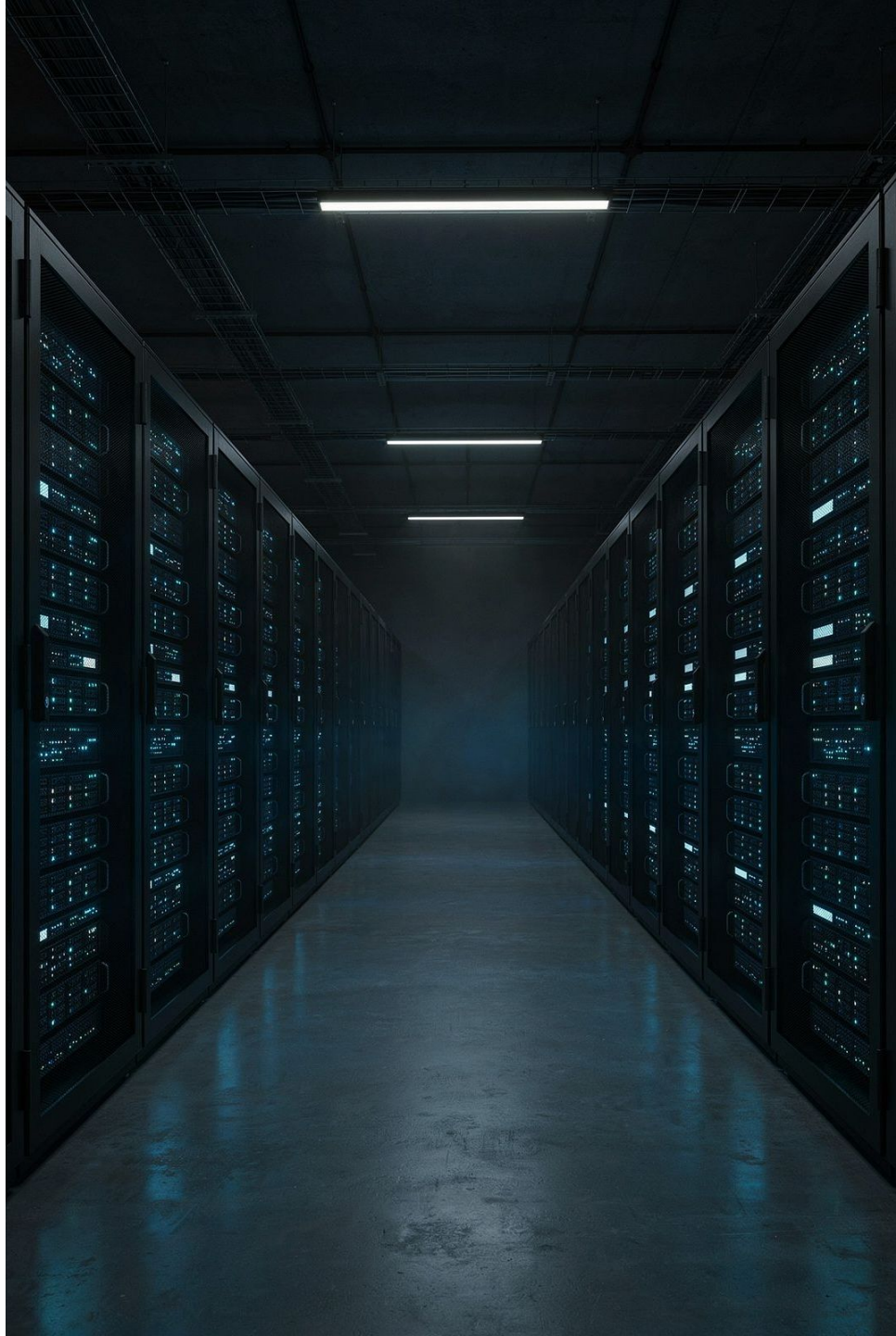
Building a Production-Grade AI Platform on AWS

A capstone journey through the full stack of AWS AI infrastructure — from GPU-accelerated model serving to serverless retrieval, automated pipelines, and enterprise-grade governance — plus the certification roadmap to prove it.

CAPSTONE BUILD · AWS AI CERTIFICATION PATHWAY



The Big Picture

Why Build a Full AI Platform?

Most teams bolt AI onto existing infrastructure as an afterthought — a single Lambda function calling an API, with no scalability, no cost controls, and no audit trail. The result: systems that work in demos but fail in production.

This capstone takes the opposite approach. Every layer of a **real-world, enterprise AI platform** is wired together intentionally — compute, retrieval, gateway, pipelines, observability, and compliance — mirroring what companies like Airbnb, Goldman Sachs, and Intuit run at scale.

What Makes It "Production-Grade"?

- Scales to zero when idle, bursts under load
- Every inference request is logged and auditable
- Models retrain automatically on new data
- Costs are visible and controllable in real time

Platform Architecture

The Eight Pillars at a Glance

Each component solves a distinct problem. Together they form a cohesive, end-to-end AI system.



Amazon Bedrock

Foundation model hub — Claude, Titan, Llama — accessed via a single managed API. No GPU provisioning required for inference.



EKS + Karpenter

Kubernetes cluster with GPU node groups that scale to zero when idle and spin up in under 90 seconds under load.



KServe

Model serving layer on Kubernetes. Canary rollouts, A/B testing, and autoscaling baked in — think of it as Kubernetes-native SageMaker endpoints.



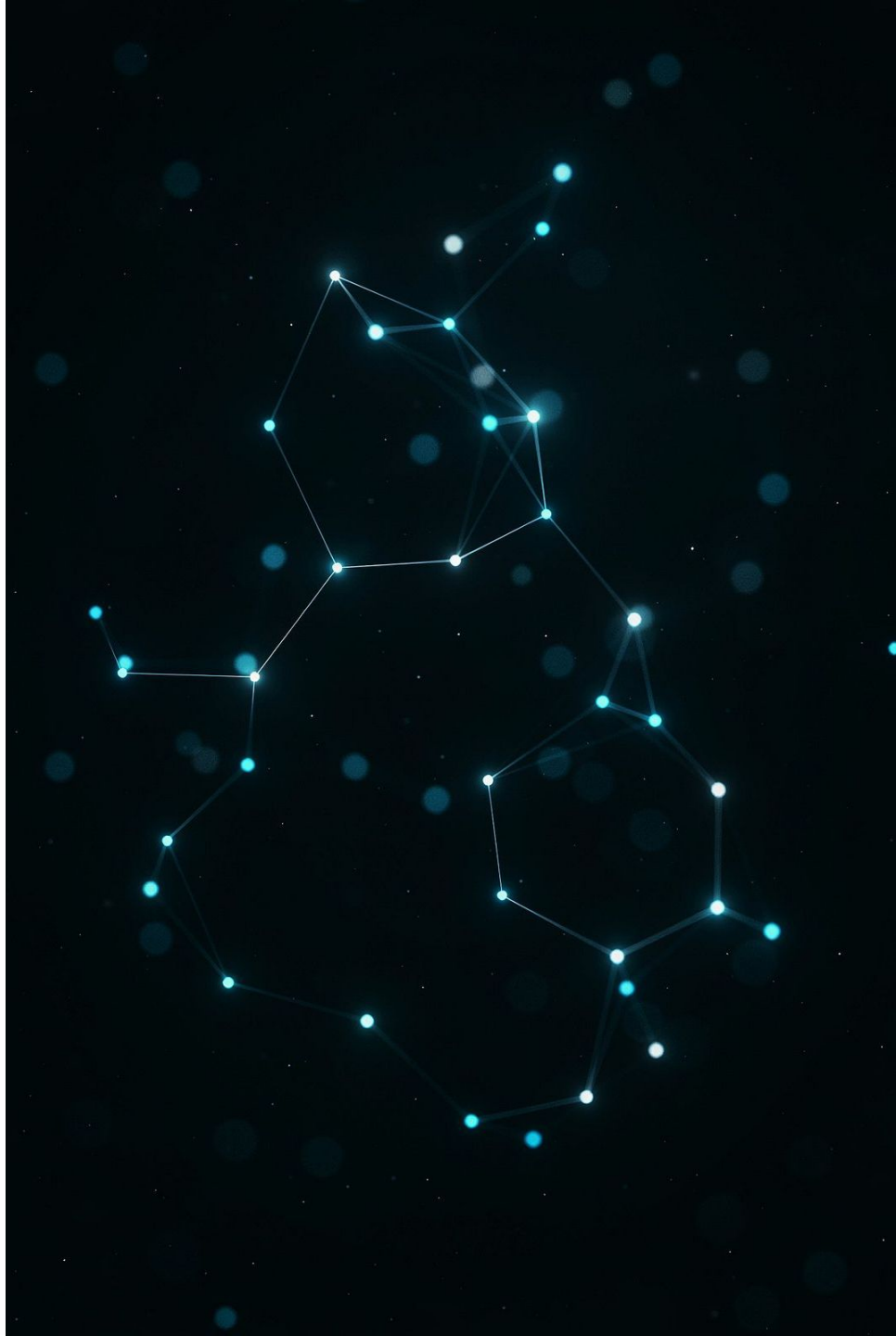
OpenSearch Serverless RAG

Vector store for Retrieval-Augmented Generation. Ground model answers in your private documents without fine-tuning.

Layer 1 — Foundation Models

Amazon Bedrock: Your Model Hub

Think of Bedrock as a **Netflix for foundation models**. Instead of downloading and hosting each model yourself, you call a single API and choose from Claude 3, Amazon Titan, Mistral, or Meta Llama — paying only per token consumed. The capstone uses Bedrock as the "brain," with every other component acting as the nervous system around it. Guardrails are configured to block prompt injection, PII leakage, and off-topic queries — critical for enterprise deployments handling sensitive data.



Layer 2 — Compute & Serving

EKS GPU Nodes + Karpenter + KServe

The Problem They Solve

Running a GPU 24/7 for a model that gets 10 requests per hour costs thousands of dollars monthly in wasted compute — the same mistake early Kubernetes adopters made with over-provisioned EC2 fleets.

Karpenter watches pending pods and provisions exactly the right GPU instance (p3, g5, or Inferentia) in real time, then terminates it when demand drops. **KServe** manages the model lifecycle on top — health checks, canary deployments, and traffic splitting between model versions.

Real-World Analogy

It works like a **ride-share fleet**: cars (GPUs) are dispatched only when passengers (inference requests) appear, not idling in a parking lot all day. A large retail company using this pattern cut GPU spend by 62% while maintaining sub-200ms p99 latency.

> Karpenter consolidation runs every 30 seconds — unused nodes are terminated automatically, keeping the bill lean.

Layer 3 — Knowledge & Retrieval

OpenSearch Serverless RAG: Grounding AI in Your Data

Large language models hallucinate when they don't know the answer. **Retrieval-Augmented Generation (RAG)** fixes this by giving the model access to a private knowledge base at query time — like handing an open-book exam instead of a closed one. The capstone ingests company documentation, support tickets, and product specs into **OpenSearch Serverless** as dense vector embeddings. At inference time, the top-k most relevant passages are retrieved and injected into the prompt, dramatically improving answer accuracy without expensive fine-tuning. Serverless means the vector index scales with query volume automatically.



Layer 4 — Gateway & Pipelines

API Gateway AI Gateway + SageMaker Pipelines

API Gateway AI Gateway

Every AI request flows through a single managed gateway that enforces **rate limits, authentication, semantic caching, and usage quotas** per API key. This prevents a single runaway application from burning through the entire monthly token budget — a real incident that cost one fintech startup $80K in a weekend.

SageMaker Pipelines

Model retraining is fully automated. When new labeled data accumulates beyond a threshold, a pipeline triggers: data validation → preprocessing → training → evaluation → conditional registration. Only models that beat the baseline metric graduate to the model registry and get promoted to production via KServe.



Layer 5 — Observability & Compliance

Cost Explorer Dashboard + CloudTrail Audit

💰 Cost Explorer Dashboard

A custom Cost Explorer dashboard tags every inference call, training job, and storage bucket by team, model, and environment. Finance can see that the fraud-detection model costs $1,200/month while the chatbot costs $400/month — enabling data-driven budget conversations instead of sticker shock at month-end.

🔍 CloudTrail Audit

Every API call — model invocation, parameter change, IAM role assumption — is logged immutably in CloudTrail and streamed to OpenSearch for near-real-time alerting. This satisfies SOC 2, HIPAA, and ISO 27001 audit requirements out of the box. If a model is ever queried with sensitive data, the full trace is preserved for forensic review.

🏗️ Full-Stack Cohesion

The dashboard correlates cost spikes with specific pipeline runs and CloudTrail events, so engineers know not just *what* was expensive, but *why* — a capability most teams only build after a painful billing surprise.

Next Chapter

AWS AI Certification Pathway

Building the platform is step one. **Validating that knowledge with AWS certifications** is step two — turning hands-on experience into industry-recognized credentials that open doors to senior ML engineering and AI architect roles.

01

AWS Certified Cloud Practitioner

Foundation. Understand core AWS services, billing, and the shared responsibility model. Ideal starting point before specializing.

02

AWS Certified Solutions Architect – Associate

Design resilient, cost-optimized architectures. Directly applicable to the EKS, VPC, and IAM patterns built in the capstone.

03

AWS Certified Machine Learning – Specialty

Deep focus on SageMaker, data engineering, model evaluation, and ML ops — the exam most directly aligned with this capstone's tech stack.

04

AWS Certified AI Practitioner

Newest credential. Covers generative AI, Bedrock, responsible AI, and prompt engineering — the emerging layer every AI engineer needs validated.

Capstone Complete

Key Takeaways & What Comes Next

What You've Built

Bedrock + KServe

Flexible model serving at any scale

RAG Pipeline

Accurate, grounded AI answers

MLOps Loop

Continuous retraining, zero downtime

Full Governance

Cost visibility + immutable audit trail

Your Next Steps

1. Register for **AWS AI Practitioner** exam — directly maps to Bedrock work done here
2. Add a **multi-region failover** pattern to the EKS cluster for 99.99% SLA
3. Instrument **model drift detection** in SageMaker Model Monitor
4. Open-source the Terraform modules to build your public portfolio

The same architecture you built here powers production AI systems at Fortune 500 companies. You didn't build a demo — you built the real thing.